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Review

Microaerobic condition as pretreatment for improving anaerobic digestion: A review

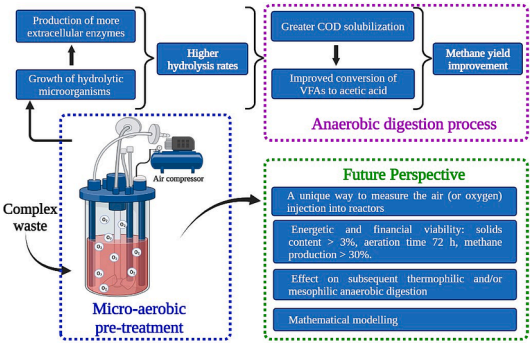
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HIGHLIGHTS

- The effect of microaerobic pretreatment (MAP) on anaerobic digestion was reviewed.
- MAP improves methane production by between 30% and 216% by hydrolysis improvement.
- A new mechanism for explaining microaerations impact on hydrolysis was proposed.
- Financial analysis shows that MAP is interesting for industrial implementation.
- Efforts on environmental and economic analysis and modelling of MAP must be done.

GRAPHICAL ABSTRACT



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ABSTRACT

Pretreatment of waste before anaerobic digestion (AD) has been extensively studied during the last decades. One of the biological pretreatments studied is the microaeration. This review examines this process, including parameters and applications to different substrates at the lab, pilot and industrial scales, to guide further improvement in large-scale applications. The underlying mechanisms of accelerating hydrolysis and its effects on microbial diversity and enzymatic production were reviewed. In addition, modelling of the process and energetic and financial analysis is presented, showing that microaerobic pretreatment is commercially attractive under certain conditions. Finally, challenges and future perspectives were also highlighted to promote the development of microaeration as a pretreatment before AD.

1. Introduction

Anaerobic digestion (AD) is a mature technique for the decomposition of organic waste materials and wastewater (Chen et al., 2020a; Wu et al., 2019). The main characteristic and advantage of this process is biogas production, considered as feedstock for synthesizing valuable chemicals and producing hydrogen (Awe et al., 2017; Tabatabaei et al., 2020). In addition, digestate left out can be utilized for soil amendment,

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fertilizers, and livestock bedding since it is devoid of harmful pathogens (Uthirakrishnan et al., 2022).

Even though AD presents these advantages, the low hydrolysis rate is a drawback in the process that has been studied during the last decades (Nguyen and Khanal, 2018). Improvement of the hydrolysis rate has received more attention because of the necessity of reduction on biodigesters and the improvement of biogas production yield. For instance, the yield of waste-activated sludge is around 0.24 L/g volatile solid (VS), while substrates with lower biodegradability (i.e., dairy cattle manure) have a value as low as 0.135 L/g volatile solid (VS) (Bremond et al., 2018), showing that process for improving these yields are necessary. Among the different pretreatment processes studied, microaerobic pretreatment has evidenced a good potential due to the hydrolytic products from the process are biodegradable and non-toxic, as opposed to thermochemical pretreatments that may produce complex wastes (Gonzalez et al., 2018). Furthermore, some studies have shown that this process can be economically feasible. Indeed, Gonzalez-Gonzalez and Cuadros (2015) demonstrated that AD of olive mill wastewater will be economically feasible if the waste is pretreated by a microaerobic process since the period of return of investment obtained under these conditions does not exceed 6 years. Furthermore, examples of industrial applications also have been presented (Bremond et al., 2018), showing that this technology is already available at an industrial scale.

There are several reviews about the effect of microaeration on AD (Chen et al., 2020a; Fu et al., 2023; Girotto et al., 2018; Nguyen and Khanal, 2018); however, these works focused on the possible effects of microaeration directly on anaerobic digesters, especially *in situ* microaeration. On the other hand, other types of pretreatments also have been reviewed, focusing on biological processes applied to lignocellulosic substrates (Bremond et al., 2018) or on specific types of substrates such as biosolids (Mitraka et al., 2022; Uthirakrishnan et al., 2022). To the author's knowledge, there is no review of the effect of microaeration as a pretreatment for improving AD. Aspects such as energy and financial analysis are still necessary to determine if this technology can be competitive in financial terms regarding other pretreatment processes. Thus, the present work was focused on the use of microaeration as a pretreatment process for AD, including aspects such as the definition of pretreatment with air, mechanisms of hydrolytic improvements and modelling of this process. A brief energetic and financial analysis is also provided to indicate which conditions can be implemented industrially.

2. Problems with the microaerobic definition and units used to measure it

As was mentioned, microaerobic pretreatment can increase the hydrolytic rate. This improves the AD kinetic, allowing smaller bioreactors and higher methane production. The definition of microaeration has been related majorly to the direct application of air inside the anaerobic bioreactor and not to the use of it as a pretreatment. In 2011, Botheju and Bakke (2011) gave a broader definition of microaeration, indicating that the terms "limited aeration", "limited oxygenation" or "partial aeration" are defined as "a process where some amount of oxygen is introduced to an anaerobic biochemical process so that there would be both the anaerobic and aerobic biological activities happening in a single bio-reactor". Later, Nguyen and Khanal (2018) indicated that microaeration is the dosing of a small amount of air or oxygen (typically ranging from 0.005 to 5 L O_2/L reactor/day) into an anaerobic system, while Chen et al. (2020a) define microaeration as a system with a low concentration of dissolved oxygen (DO) between aerobic and anaerobic environments. This system is constructed by dosing a small quantity of air or oxygen (DO range of 0.1–1 mg/L) into the anaerobic bioreactor. Finally, Zhang et al. (2021) defined microaeration as "an environmental condition with a low dissolved oxygen concentration, specifically between an aerobic and anaerobic environment, that is obtained by injecting a trace amount of air or oxygen into an anaerobic bioreactor (0.1 – 1 mg/L)". As can be seen, the definition is always regarding using an airflow inside the anaerobic reactor, not in another unit operation such as a pretreatment process. Thus, there is no clear definition of microaeration as the pretreatment of organic waste. We proposed here a definition of microaeration applied as a pretreatment, based on Chen et al. (2020a) definition: *Microaeration as pretreatment is the dosing of small quantities of air or oxygen (DO range of 0.1–1 mg/L) into a pretreatment reactor before AD to obtain a system with a low concentration of dissolved oxygen (DO) between aerobic and anaerobic environments*. This definition clarified two important aspects of microaeration as pretreatment: It is applied in other unit operations and the DO concentration can have values up to 1 mg/L in the liquid phase.

On the other hand, microaeration intensity (or rate), which is measured in terms of aeration or oxygen flow rate, has different units for expressing it, using mainly the volume of air/volume of liquid min (vvm (L_{air}/L_{liquid} min) and the volume of air/mass of total solids min (vmm, L_{air}/g Total Solids (TS) min), as is shown in Table 1. Different aeration rate units are a problem for comparing the results of different work groups and involve difficulties in scaling up the pretreatment process. On the other hand, variety or waste typologies make it impossible to have one single unit to express micro aeration. Thus, a standard unit, depending on the type of waste (liquid or solid) for the range of specific aeration, is necessary. Here we proposed that the best units for the aeration rate are the vvm for conventional AD (organic waste with a low total solids (TS) content, i.e., below 10%) and vmm for semi-dry (10% < TS < 15%) and dry (≥15% TS) AD, due to the following reasons:

a. Easier to implement at an industrial scale: For conventional AD, the volume of liquid is typically kept without change during the operation of the process. Thus, the air injected into the pretreatment can be easily calculated using the liquid volume as the base. For semi-dry and dry AD, the easier way to control and calculate the aeration is with respect to the solid load, as the management of this type of operation depends on the solid content.
b. Easier to compare between other plants or scales: These parameters (vvm and vmm) could be easily compared not only between research groups but also between industrial plants that use conventional AD (vvm) or dry or semi-dry AD (vmm).
c. Standard parameter in other bioprocesses: Because the vvm is also used in other types of bioreactors, using this unit for conventional AD also can improve the understanding of the conditions for other professionals.

Another aspect that must be analyzed is the dosing method, as shown in Table 1. Various aeration modalities have been performed by air injection, oxygen injection or even natural aeration, where the tested configuration is left open with or without shaking (Chen et al., 2020a). In addition, the aeration frequencies (or dosing) can be one time, intermittently or continuously, depending on practical requirements (Chen et al., 2020a). According to Table 1, the most used dosing is continuous aeration, although using different time ranges, followed by pulse injection in batch systems. Continuously forced aeration has the advantage of easier implementation at a higher scale (pilot and industrial scale) and easier control of DO in the system, so it should be the way to work on an industrial scale. Natural aeration can be impractical at an industrial scale because of the waste volume involved in solid or water treatment. In any case, the aeration rate should be carefully considered because the excess oxygen amount reduces the methane potential, while insufficient oxygen amount potentially fails to produce any effect (Chen et al., 2020a; Girotto et al., 2018) as shown in several studies.

Another aspect that required mention is the hydraulic retention time (HRT) in the microaerobic pretreatment reactor. Similar to the aeration rate, HRT depends on the type of waste (liquid or solid), being higher in the pretreatment of solid waste. According to the studies that used liquid wastes (primary and mixed sludge, sewage sludge, olive mill wastewater, and blackwater, among others), HRT should be between 1 and 6

Table 1
Summary of different units used in the literature for specific aeration rates and dosing methods applied in microaeration as pretreatment.

Airflow rate unit	Dosing method	Reference
vvm	Continuous, 24 h	Hasegawa et al. (2000)
mL/d	Intermittently 15 s every minute	Johansen and Bakke (2006)
L/min	Continuous, between 5 min every 24 h to 1 h every 24 h.	Zhu et al. (2009)
vvm	Continuously	Diak et al. (2013)
ml O_2/L $_{liquid}$ d	Continuous, 4 days	Lim and Wang (2013)
vvm (mL O_2/L d)	Single dosing, 1–2 h after feeding.	Lim et al. (2014)
vmm (L_{air}/kg TS d)	Semi-continuous time based, 17 days	Xu et al. (2014)
mL O_2/g VS $_{substrate}$	Pulse Injection	Fu et al. (2015)
vvm (L_{air}/L $_{reactor}$ min)	Continuous, between 5 and 7 days	Gonzalez-Gonzalez and Cuadros (2015)
vvd (mL_{air}/L $_{liquid}$ d)	Pulse Injection	Fu et al. (2016)
vvm (L_{air}/L $_{liquid}$ min)	Continuous, 24–96 h	Montalvo et al. (2016a)
vvm (L_{air}/L $_{liquid}$ min)	Continuous, 24–60 h	Montalvo et al. (2016b)
L/h	24 h, continuously	Rafieenia et al. (2017)
vmm (L_{air}/kg $_{substrate}$ min)	Continuous, 8 days	Zhou et al. (2017)
vmm (L_{air}/min kg $_{waste}$)	Continuous, 4–14 days	Ni et al. (2017)
mL O_2/g VS $_{added}$	Pulse injection	Sawatdeenarunat et al. (2017)
mL O_2/g VS $_{added}$	Pulse injection	Tsapekos et al. (2017)
—	Open to the air	Duarte et al. (2018)
vmh (L_{air}/g $_{solid}$ h)	Continuous, 1–4 days	Fisgativa et al. (2018)
vvm	Continuous, 24 h	Hou et al. (2018)
vvm	Continuous, 1 day	Sahu et al. (2019)
vvm	Continuous, 1 day	Nguyen et al. (2019)
vmm (Lair/g TS min)	Continuous, 4 h	Ruan et al. (2019)
vvm	Continuous, 2–10 days	Alrawashdeh and Al-Essa (2020)
vmd (L_{air}/kg TS d)	15 min every two hours	Bacab et al. (2020)
Open to the air	Continuous, 12 h	Chen et al. (2020b)
vmh (L_{air}/g $_{solid}$ h)	Continuous, 2 days	Fisgativa et al. (2020)
vvm	Continuous, 20–60 h	Rashvanlou et al. (2020)
vvm	10 times daily for 15 min	Rubezius et al. (2020)
mL O_2/g VS d	Continuous, 10 days	Wang et al. (2020)
mg O_2/L $_{reactor}$ cycle	Pulse injection	Yu et al. (2020)
mL O_2	Pulse injection	Zhen et al. (2020)
Oxidation-reduction potential ORP (−420 mV)	Controlled at values for obtaining −420 mV	Chetawan et al. (2021)
—	Open to the air	Xu et al. (2021)
vvm	Continuous, 20–60 h	Rashvanlou et al. (2021)
mL O_2/g VS $_{added}$	Pulse injection	Song et al. (2021)
mg O_2/g COD per cycle	Pulse injection	Guo et al. (2021)
—	Open to the air, 6 days	Cheng et al. (2021)
vmm (L_{air}/kg min)	Continuous, 14 d	Xiang et al. (2022)
—	Open to the air, 10 days	Karami et al. (2022)
mL O_2/mL min	continuous	Zhu et al. (2022b)
mL/g VS d	Pulse injection	Zhu et al. (2022a)
vvm	intermittent microaeration for 5 min	Cao et al. (2022)
vvm	Continuous, 2 days	Castillo et al. (2022)
N mL air/d	Pulse injection	Zhan et al. (2023)

days. In the case of solid waste (Municipal solid waste (MSW), food waste, and lignocellulosic waste, among others), HRT varied between 6 and 14 days, depending on the solid load. In both cases, range times improve the hydrolytic rate and improve biogas production.

3. Application of micro aeration as a pretreatment method

3.1. Effect on different substrates

Several studies have investigated the effect of microaerobic pretreatment at different substrates and its effect on the hydrolysis phase and subsequent methane production in AD. A resume of it is shown in the supplementary material. As different feedstock has different characteristics and challenges, the goal of the microaerobic pretreatment depends on the substrate used. Microaerobic pretreatment can break recalcitrant compounds to increase biodegradability for solid or lignocellulose waste. According to Zhang et al. (2021), micro-aeration conditions enrich the facultative hydrolytic and acid-producing bacteria in the system, which leads to an increase in the production of high amounts of extracellular enzymes, accelerating the hydrolysis of refractory organic matter and thus improving VFA production.

For many substrates, hydrolysis can hinder the AD process in its initial phase and some studies have proved that little amount of air can improves the hydrolysis process with further successful results in AD (Fisgativa et al., 2018; Fu et al., 2015; Hou et al., 2018; Montalvo et al., 2016a; Ruan et al., 2019; Tsapekos et al., 2017; Xu et al., 2014; Zhou et al., 2017). For lignocellulosic substrates, several works have been proposed. Zhu et al. (2009) working with vegetable and flower wastes in a hydrolysis reactor before a methanogenic bioreactor and including recirculation from the anaerobic to hydrolytic reactor found that sufficient microaeration (5 min of aeration every 12, 4 or 1 h) promoted the hydrolysis of easily biodegradable carbohydrates and proteins during the early period. However, the microbial activity under micro-oxic conditions was impaired by liquid recirculation with methanogenic effluent. Therefore, a sufficient microaeration strategy should be employed only during the early digestion period, to enhance the hydrolysis of easily biodegradable organics, promote acidogenesis and avoid the accumulation of intermediates such as lactic acid. Lower aeration rates (5 min of aeration every 24 h) must be avoided because this condition had the lowest enzymatic activities, indicating that it had the least appropriate conditions for anaerobes. Later, Fu et al. (2015) studied the effect of thermophilic microaerobic pretreatment (TMP) on the AD of corn straw in batch systems, showing that the maximum methane yield was 16% higher than that of an untreated group using an aeration rate of 5 mL/g VS $_{substrate}$, because of the improvement of the hydrolysis efficiency. Furthermore, the microaeration reduced the lag-phase time. Similar results were presented by Xu et al. (2021), also using corn straw as substrate but at mesophilic temperature (35 °C), who indicated that the maximum methane yield potential was 17% higher than that of the control. Furthermore, the total solid reduction was greatly affected by pH, with the highest hydrolysis rate obtained at pH = 8, i.e., alkaline pH improved the hydrolysis of cellulosic waste in batch systems. Zhou et al. (2017), using rice straw as substrate, obtained an improvement in the hydrolysis phase when a continuous air flow rate of 0.5 L/kg $_{substrate}$ min was applied for 2 days at 35 °C. The hydrolysis efficiency increases up to ~ 12% leading to improve methane production (i.e., 16% higher than the control reactor) for 2 days' aeration. When aeration time rises from 4 to 8 days, the methane yield decreases under the methane yield of the untreated rice straw, tallied to the reduction of cellulose degradation. As stated, more abundance of hydrolytic bacteria may be present in the microaeration environment which favours the secretion and activity of hydrolytic exo-enzymes such as cellulases, xylanase, and lipases for efficiency biodegradation (Bacab et al., 2020; Duarte et al., 2018; Nguyen et al., 2019).

Other more biodegradable wastes, such as sewage sludge, have also been studied. Microaerobic hydrolysis pretreatment was employed for mixed sludge (i.e., 60% primary sludge + 40% waste-activated sludge) at 35 °C applying an aeration rate between 4 and 48 h at 1 to 6 air vmm (L_{air}/g TS min) (Ruan et al., 2019). Best results were obtained for 4 vmm where proteins, polysaccharides, and total organic carbon concentrations reached their maximum values in agreement with the higher activity of protease (4.01 mU/g VS) and α-glucosidase (4770 mU/g VS). Total volatile fatty acids (VFAs) production showed the same trend as released dissolved organic matter in the hydrolysis of proteins and polysaccharides. These effects in the hydrolysis phase were further confirmed by its impact on the AD process, in which methane yield reached its maximum value (i.e., 181 mL/g VS (4vmm) vs. 149 mL/g VS (0 vmm)). Similar findings were noticed by Johansen and Bakke (2006)

where higher hydrolysis efficiency was achieved for proteins and carbohydrates compared to those obtained for lipids hydrolysis when ~ 500 mL_{air} per day was intermittently applied 15 s every minute into a primary sludge. Diak et al. (2013) showed efficient protein degradation, organic matter solubilization and reduction in the total and volatile solids for microaerobic pretreatment of primary sludge when applying 0.00156 vvm of air, but no improvement in the hydrolysis of carbohydrates was observed. The combined effect of several factors such as airflow rate (i.e., 0.1, 0.3, 0.5, and 1 vvm), pretreatment time (i.e., 24, 48, and 60 h) and temperature (i.e., 20, 25 and 35 °C) on the microaerobic pretreatment efficiency and methane generation of AD of sewage sludge was studied by Montalvo et al. (2016a). It was found that microaerobic pretreatment improved the hydrolytic process, with an increase of proteins, total sugars, sCOD, and greater removal of volatile suspended solids compared to a system without aeration. The best conditions were found at 0.3 vvm, 48 h of aeration time, and a temperature of 35 °C. Under these conditions, methane production increased by 211% compared to the non-aerated treatment. Hasegawa et al. (2000) applied microaerobic pretreatment to organic sludge (at the thermophilic condition) and found that microaerobic levels of oxygen (i.e., 0.08–0.4 vvm) reduced the VFAs accumulation, causing higher biogas production (150%) compared to the not pretreated sample. Other researchers pre-treating sludge found similar results when applying different microaeration strategies (Hou et al., 2018; Montalvo et al., 2016b; Rashvanlou et al., 2020). For example, Rashvanlou et al. (2021) applied a combined pretreatment strategy for sewage sludge, using low-frequency ultrasonic and microaerobic pretreatment. The highest sludge solubilization was observed at 0.2 vvm and 40 h, reaching up to 65% soluble chemical oxygen demand (sCOD) increasing. They found a slight increase in the enzymatic activity, but no significant change in lipase activity as previously reported by Johansen and Bakke (2006) for lipids degradation. When Rashvanlou et al. (2021) applied only microaerobic pretreatment, the methane yield increased by 148%. Nevertheless, the combined ultrasonic and microaerobic pretreatment increases the solubilization (up to 70%), the microbial activity (2080%) and the biogas production (193%) compared to each individual pretreatment. For the same substrate (i.e., municipal sludge with 70% secondary and 30% primary sludge), pretreatment between 0.1–0.5 vvm and 20–60 h improved the organic matter hydrolysis rate between 41% and 65%, with a volatile suspended solids (VSS) reduction of 31% for 40 h (Rashvanlou et al., 2020). Using a microaerobic pretreatment of 0.2 vvm by 48 h, the methane yield increased by 221%, in agreement with results found by Montalvo et al. (2016a) where methane yield increased at similar values (i.e., 211%) after microaerobic pretreatment. The impact of microaerobic pretreatment (0.35 vvm by 48 h and 35 °C) on the waste-activated sludge was also investigated for long-term AD (Castillo et al., 2022). The data showed an increase of 30% in soluble proteins and sCOD, with better performance for the digester fed with pretreated sludge compared to the digester fed without sludge microaerobically pretreated, in a period of 95 days. The positive effect of microaerobic pretreatment during the semi-continuous operation was also confirmed by results previously obtained by Gonzalez-Gonzalez and Cuadros (2015) when treating a different substrate, such as olive mill wastewater. When 0.65 $L_{air}/L_{reactor}$/min was applied to the olive mill wastewater (OMWW) at 5- and 7-days' aeration, it was possible to reduce the content of polyphenols up to 90% with a reduction of the initial organic matter content of 21%. The performance of the AD in semicontinuous mode (i.e., for 80 days) showed the best results with no inhibition and stable process for the 5 days' aeration pretreatment in comparison with the control reactor and the one pretreated for 7 days. The anaerobic performance in semi-continuous mode (40 days) of the same substrate (i.e., olive mill wastewater) was also investigated after applying 0.5 $L_{air}/L_{substrate}$/min during 2-, 4-, and 6-days' aeration (Alrawashdeh and Al-Essa, 2020). The olive mill wastewater (OMW) was aerobically pretreated and co-digested with activated sludge (AS) for AD performance. The best results were obtained for 4 days- of pretreated OMW, where the highest removal of polyphenols (60%), volatile solids (VS) (56%), and total COD (tCOD) (70%) was attained. The methane content in the biogas increased from 25.8% for the untreated mixture (i.e., AS + non-treated OMW) to 68% for the treated mixture (i.e., AS + 4 days-pretreated OMW), with the highest biogas yield (0.44 Nm^3/kgVS vs. 0.107 Nm^3/kg VS). As can be seen, there are few studies applying microaerobic pretreatment in continuous or semi-continuous AD processes, therefore, more studies are necessary to compare and determine the real effect of microaerobic pretreatment in continuous processes.

The role of microaerobic pretreatment on substrates with higher biodegradability, such as food waste and wastewater, has also been investigated (Fisgativa et al., 2020; Fisgativa et al., 2018; Lim et al., 2014; Lim and Wang, 2013; Rafieenia et al., 2017; Sahu et al., 2019). (Lim and Wang, 2013) pretreated a mixture of food waste (150 g) and brown water (2 L) applying 37.5 mL O_2/L_{liquid} d for 4 consecutive. They show that, besides higher solubilization of COD, microaerobic pretreatment led to the enhancement of short-chain fatty acids conversion to acetic acid, entailing a methane yield improvement between 10 and 21%. This was also supported by Lim et al. (2014) where a more diverse bacterial community and higher VFAs production were achieved when the same substrate (mixing of food waste and brown water) was treated. Similar results were found by Xu et al. (2014) treating food waste in a two-phase anaerobic digester, where adequate microaeration time (i.e., 2 and 3 h) enhanced the proteins and carbohydrates hydrolysis by 38–64% and 21–27%, respectively. The best result was obtained for 3 h of aeration time and 258 L_{air}/kg TS/d, resulting in the maximum methane yield in the methanogens phase of a UASB reactor. Rafieenia et al. (2017) likewise observed that aerated proteins-rich substrates (i.e., synthetic food waste) yield the best results among carbohydrates or lipid-rich material for methane production. Zhou et al. (2017) complemented all these results, indicating that higher aeration intensity disfavors methane production due to the oxidation of intermediate soluble products, which decreases the solubilization of COD. Dry AD of food waste was also investigated after different aerobic pretreatment conditions (i.e., dynamic aeration-39 °C-2 day-50 L_{air}/h vs. static storage-room temperature-2 days) and leachate recirculation (Fisgativa et al., 2020). No results were reported about the effect of aerobic pretreatment in the organic matter solubilization before dry-AD. The solid-state AD performance of pretreated food waste showed that the 2 days storage state did not improve the methane production and rapid acidification was observed due to an imbalance of acids production and buffer capacity. Nevertheless, 2 days of dynamic pre-aeration stabilized the acids production along the digestion process leading to an enhancement of the methane production. During a short period of air in solid-state fermentation, the biowaste undergoes partial decomposition with enzyme stimulation, which prepares the waste for subsequent AD. This technique is already applied commercially by the company Zero Waste Energy LLC (USA) under the brand named Smartferm®. A similar substrate (i.e. kitchen waste) was also oxygen-pretreated at the start of the experiments to evaluate its impact on the anaerobic fermentation using a batch reactor (Zhen et al., 2020). Better decomposition of small organic and nitrogenous molecules was obtained when 20 mL O_2 was supplied, in comparison to the rest of the oxygen supplementation dosages (i.e., 5, 10, 15 mL O_2). This study suggests that facultative anaerobes improved their capacity to produce and utilize nitrogenous compounds during fermentation at higher O_2 doses. Sahu et al. (2019) also revealed a reduction in the ammonia accumulation levels inside the digester, with better process stability when kitchen waste was microaerobically pre-treated (i.e., 0.19–0.22 vvm). Cheng et al. (2021) worked on the pretreatment of food waste (FW) and sewage sludge (SS) before a semi-continuous co-digestion process. They found that aerobic pretreatment increased the concentration of ammonia nitrogen in the substrate, effectively buffering the acid produced and promoting CH_4 production. Indeed, the co-digestion of pre-treated FW-SS had a CH_4 yield 54% higher than the system without pretreatment, working at

OLR = 4 g VS/L d and HRT = 16 days. This work was one of the first to study the pretreatment for co-digestion and demonstrate the process in semi-continuous operation, a situation closer to an industrial bioreactor.

Finally, agro-industrial waste, such as manure from different industries, has been used for microaerobic pretreatment. Chetawan et al. (2021) proposed a pretreatment including the recirculation of sludge from the anaerobic plug-flow reactor to the CSTR microaerobic pretreatment reactor, using pig manure as substrate. This research proposed the use of ORP as a variable to control to maintain the microaerobic bioreactor, showing that this parameter was an effective means to regulate fermentative reactions. Indeed, in continuous digester operations using piggery wastewater, a combination of ORP control, mesophilic temperature (30 °C) and returned sludge increased methane yield as high as 20%, mainly to the enlarging hydrolytic bacteria abundance in the sub-thermophilic reactor use for pretreatment. Two aspects can be highlighted in this work: (i) the continuous operation of the system pretreatment/AD and (ii) a preliminary cost-benefit evaluation. The first point is important because it shows that in an operation similar to the industrial one, the performance of a microaerobic pretreatment allows a permanent increase in methane yield. Furthermore, the economic study demonstrated that the payback period is approximately 3.88 years for a project that included the pretreatment operations for 450 m^3/d on average piggery wastewater with a total solid of 1%.

Rubezius et al. (2020) studied the effect of an aerobic pretreatment of poultry manure on the biogas production process and the reduction of ammonia emissions using a batch system. This work has been the only one that reported the effect of microaerobic pretreatment on NH_3 emissions, a common problem in manure AD. The results of this work showed that the microaerobic pretreatment of poultry manure has a positive impact on biogas yield, biogas quality and substrate degradation rate. The concentration of methane in biogas after pretreatment was influenced by the aerobic process and increased from 55% to 63% while activating the ammonification process in the substrate. Indeed, in all pretreatments, substrate NH_4^+-N increases more than 2-fold. (Cao et al., 2022) studied the roles of microaeration on mono-digestion of swine manure (SM) for VFAs production and co-digestion of SM with corn silage (CS) for lactic acid (LA) production. This study revealed that microaeration significantly enhanced VFAs and LA production from mono-digestion of swine manure and co-digestion of swine manure with corn silage, increasing the maximum VFAs concentration by 20% and shortening the time to reach the maximum from 18 days to 10 days. Furthermore, the conversion of LA to VFAs was effectively limited, achieving a higher LA concentration of 26.08 g COD/L.

As can be seen from the literature analysis, microaerobic pretreatment has been extensively studied for different types of substrates at the lab and industrial scale. In all the experiences, a good pretreatment performance was observed, in batch and continuous or semi-continuous operation. Most of the studies were performed in batch mode, which does not necessarily represent the industrial operation of pretreatment and AD. Although some industrial applications have been reported, more studies in continuous or semi-continuous conditions are still essential to determine the real performance of the pretreatment in similar conditions to the industrial application of AD. Another important aspect is that most studies have been conducted under mesophilic conditions. However, temperature can influence numerous aspects such as microorganism activity, substrate biodegradation, and enzyme production. In this sense, it is known that higher temperatures can accelerate biochemical reactions and increase substrate degradation rates. Nevertheless, it is important to consider that each substrate and microorganism may have specific temperature requirements. Therefore, to maximize the efficiency of microaerobic pretreatment, it is essential to conduct further studies to determine the optimal temperature for the pretreatment of each substrate.

3.2. Effect of microaeration on microorganisms and mechanisms behind the process

The improvement in the hydrolysis process by DO presence in the system is explained by the different microorganisms favoured by oxygen. In this sense, several studies in both lab and pilot scales indicated that *Firmicutes*, *Bacteroidetes*, and *Proteobacteria* phylum are responsible for higher hydrolysis rates (see supplementary material). Lim et al. (2014) showed that the microaeration increases in 0.24 and 24-fold the *Firmicutes* and *Bacteroidetes* phylum, respectively, with respect to an anaerobic reactor, while Fu et al. (2016) indicated that the microaeration increases in 0.06-fold the *Firmicutes* phylum regarding a strict anaerobic reactor. Cao et al. (2022) working on the VFA-increasing in the AD of swine manure using batch reactors, indicated that the growth of *Clostridium* and *Terrisporobacter* was more salient under microaerobic conditions. Chetawan et al. (2021) using a CSTR as a microaerobic prehydrolysis system for piggery wastewater, indicated that the use of a pre-treated substrate in a plug-flow anaerobic digester produced an expansion in the hydrolytic group presents in the anaerobic digester, being the hydrolytic bacteria *Clostridium* spp. and *Thermomonas* spp. which had the highest development. It is important to note that temperature also can explain this behaviour in the work of Chetawan et al. (2021). Duarte et al. (2018) indicated that microaeration promoted slightly higher oxidation–reduction potential (ORP) conditions (−250 mV vs. −350 mV), that stimulated the presence of facultative anaerobic bacteria particularly, *Proteobacteria* such as *Pseudomonas* spp. Fisgativa et al. (2018) investigated the change, in a microbial community when food waste is pre-treated microaerobically, indicating that an aerobic pretreatment of 4 days increases the presence of *Proteobacteria* (*Pseudomonas, Enterobacteriaceae* and *Achromobacter*) in the microaerobic reactor in comparison to the system without air. Guo et al. (2021) working with anaerobic sequencing batch reactors (AnSBRs) and using blackwater as substrate, obtained the same tendency in the anaerobic reactors., i.e., *Bacteroidetes* was the most prevalent phylum in the low-oxygen-dosage reactors, followed by *Proteobacteria* and *Firmicutes*. Two characteristics have the work of Guo et al. (2021): a.- They add the oxygen at the beginning of each AnSBR cycle and b.- They use several HRT to see the behaviour under different conditions. Thus, the results presented by Guo et al. (2021) can give closer ideas about the effect of microaeration as pretreatment.

Recently, Li et al. (2022) used a new concept for improving AD using side-stream microaeration (SMA), where microaeration can be operated in a side-stream reactor with a much smaller volume, which avoids heterogeneous diffusion of air in the main digester and lowers the potential risks in systems reforming and subsequent operation. They indicated that including an SMA enhanced the dominant species of hydrolytic acidifying bacteria in AD such as *Firmicutes and Actinobacteriota.* In addition, SMA increased the relative abundance of archaea such as *Methanosaeta*. The improved relative abundance of these microorganisms showed cooperation among hydrolysis, fermentation and methanogenesis. Ruan et al. (2019) using a mixed sewage sludge pre-treated microaerobically and fed to a semi-continuous reactor, showed that the effect on AD of microaerobic pretreatment is the increasing of *Proteobacteria*, *Chloroflexi*, *Bacteroidetes*, *Parcubacteria*, *Synergistetes*, *Actinobacteria* and *Firmicutes* concerning to the system fed with raw mixed sludge. This phylum is important in anaerobic hydrolysis and acidification. Song et al. (2021) using microaerobic pretreatment of paper waste, indicated that *Bacteroides* and *Macellibacteroides* were enriched after microaerobic pretreatment using sheep manure as a microorganism source, which all contributable to the cellulose degradation. Furthermore, by comparing the groups with the same microbial agent, it could be found that changing the oxygen loading might result in the alternation between hydrogenotrophic and acetoclastic methanogens, which possibly affects the methanogenesis stage. Xiang et al. (2022) used microaerobic pretreatment for the AD of rice straw and pig manure. They indicated that the microaerobic process improved the ability of

microorganisms to degrade the substrate and hydrolyze big particle-size lignocellulose, with *Candida*, *Lactobacillus*, *Bifidobacterium*, and *Acetobacter* playing major roles in degrading insoluble substrates and accumulation of VFAs. Xu et al. (2021) using a microaerobic pretreatment of corn stover showed that applying microaerobic pretreatment increased the percentage of bacterial population of the phylum *Firmicutes*, *Bacteroidota*, and *Proteobacteria*. Yu et al. (2020) studying the microaerobic pretreatment of blackwater, found that the hydrolytic/fermentative bacteria and acetoclastic methanogens (e.g., *Methanosaeta*) were in higher abundances in low-dose microaeration reactors (5 and 10 mg O_2/L $_{reactor}$/cycle), which facilitated syntrophic interactions between microorganisms. The relative abundance of oxygen-tolerant methanogens, such as *Methanosarcina,* greatly increased (from 2% to 11%) after oxygen injection. A similar finding was indicated by Cheng et al. (2021), who showed that methanogens in the aerobic pretreatment group were mainly *Methanosarcina*, while methanogens in the non-pretreatment group were mainly *Methanosarcina* and *Methanobacterium*. As a result, the main pathway in pre-aerated arrays was acetotrophy for methanogenesis and methanogenic performance for the non-pretreatment group was weaker than the pretreatment group. Zhu et al. (2022a) using a batch test with corn straw as substrate, indicated a higher microbial diversity under microaeration conditions, where the top four dominant phyla were *Bacteroidota*, *Firmicutes*, *Synergistota,* and *Spirochaetota* in that order.

As can be seen, depending on the microaerobic conditions, different types of microorganisms are stimulated, with a higher presence of hydrolytic and acidogenic biomass (*Firmicutes, Bacteroidetes* and *Proteobacteria*) as well as of oxygen-tolerant methanogen such as *Methanosarcina*. More studies can be done to dilucidated the type of microorganism depending on substrate types.

As mentioned earlier, several works have demonstrated that microaerobic pretreatment improves the AD of different types of substrates. However, the mechanisms behind the improvement have been scarcely reported. In this regard, a mechanism that could explain the practical effect of air on hydrolysis was presented by Zhang et al. (2021). According to this mechanism, under microaeration conditions, the bubbles (air) with negative charges attract nicotinamide adenine dinucleotide (NAD^+) transfer, causing an increase in the production of the reduced form of nicotinamide adenine dinucleotide (NADH). Thus, to maintain redox balance (NADH/NAD^+), NADH is regenerated from the production of fermentation intermediates, such as VFAs. For VFAs production, the hydrolysis rate should be increased by facultative bacteria to obtain the VFAs necessary for the redox balance. This facultative bacteria activity improves the electron transfer activity of microbial metabolism Zhang et al. (2021). On the other hand, the reduction of oxygen with NADH can release a large amount of energy, which further generates a proton gradient (ΔH^+), thereby improving the efficiency of hydrolysis acidification. Even though this mechanism explains the improvement in hydrolysis based on the microorganism metabolism, it does not incorporate the enzyme effects on the process. As was mentioned earlier, the improvement in the hydrolysis rate can be a consequence of enzymes generated by hydrolytic microorganisms. In this sense, recently Zhu et al. (2022a), proposed a mechanism to explain why the microaeration improved the hydrolysis and methanogenesis of AD from corn straw in batch systems with a small injection (pulse) of air, based on their observations. The mechanism proposes that oxygen introduced into the anaerobic reactor could stimulate the growth of facultative microbes (such as phylum *Firmicutes*) that, in turn, produced more extracellular enzymes (e.g., cellulase), enhancing the hydrolysis of corn straw and producing more VFAs and Soluble Humic Acids (SHAs). Then, microorganisms such as *Spirochaetaceae* (phylum *Spirochaetota*) and *Synergistaceae* (phylum *Synergistota*) oxidized VFA (acetate) into H_2 and CO_2 and provided substrates (i.e., H_2 and CO_2) for the hydrogenotrophic methanogenesis. On the other hand, this work shows that the direct-interspecies-electron-transfer (DIET) during AD was also enhanced, thus improving the methane yield.

The main difference between the two models proposed is the action of enzymes. For Zhang et al. (2021), the effect is based on the metabolism of microorganisms, while Zhu et al. (2022a) proposed a mechanism that involves the release of extracellular enzymes of some type of microorganisms and the DIET improvement. Thus, the mechanism proposed by Zhang et al. (2021) appears more general and independent of the type of microorganism, although the mechanism proposed by Zhu et al. (2022) is based on experimental results from several works that observed the increase in extracellular enzymes in microaeration environments.

Thus, to improve the possible mechanisms that explain the microaeration effect, this work proposes the scheme that appears in Fig. 1, which includes enzyme action in the mechanism proposed previously by Zhang et al. (2021). Basically, the bubbles (air) injected have negative charges that attract NAD^+ transfer, causing an increase in NADH production. To maintain redox balance NADH/NAD^+, NADH is regenerated from fermentation intermediates, such as VFAs. For VFAs production, microorganisms generate more extracellular enzymes, enhancing the hydrolysis of organic matter and producing the VFAs necessary for the redox balance. Finally, oxygen reduction with NADH releases energy which further generates a proton gradient (ΔH^+), thereby improving the efficiency of hydrolysis acidification.

3.3. Effect on hydrolytic enzymes

Hydrolytic enzymes are responsible for breaking down complex organic molecules into smaller molecules that can easily metabolize by microorganisms. Some studies have shown that micro aeration pretreatment has a positive effect on boosting the activity of hydrolytic enzymes in waste, which in turn increases the rate of organic matter degradation (Charles et al., 2009; Fisgativa et al., 2018; Rashvanlou et al., 2021; Ruan et al., 2019). This is attributed to the creation of aerobic conditions in the waste, which promotes the growth and activity of aerobic microorganisms that produce and secrete key hydrolytic exoenzymes such as proteases, amylases, lipases and cellulases. Table 2 shows a summary of different enzymatic activities reported in the literature with and without micro-aeration pretreatment. As observed enzymatic activity increased in all studies after micro-aerobic pretreatment using different waste as substrates. Fisgativa et al. (2018) detected a progressive increase in the production of exoenzymes at a higher concentration of oxygen (21% O_2) in comparison with lower aerobic conditions (5% O_2) or not aerobic conditions (0% O_2) when applied to food waste. These findings indicate that a short aerobic pre-treatment in the food waste causes a release of protein and carbohydrate-degrading enzymes such as Phosphatases, Amino-peptidases, ß-galactosidase, α - and ß- glucosidases and N-acetyl-ß-glucosaminidase which may support the further improvement of the hydrolysis phase in the anaerobic process. The authors ascribe the results to the diversification of the microbial community in waste. Charles et al. (2009) also found higher cellulose and protease activities during the pre-aeration (5 days) of the organic fraction of municipal solid waste. Ruan et al. (2019) proved the positive effect of micro-aeration as a pretreatment of mixed sludge for the improvement of enzymatic activity. At 4 air volumes per gram TS per minute, the highest protease and α-glucosidase activity was observed, favouring the hydrolysis of proteins and polysaccharides in the sludge. This result supported the further AD process in which higher acetate concentration and methane yield was obtained for the corresponding aeration rate (4 vmm) with the best enzymatic activity. These results support the idea that micro aeration is in fact a way to obtain and stimulate enzymatic activity by the promotion of the growth and metabolism of facultative bacteria in the waste. Further technical and economic analysis should be performed to evaluate the energy required to provide a small amount of air in comparison with the addition of exogenous enzymes, as the effectiveness of the process will depend on the specific characteristic of the waste.

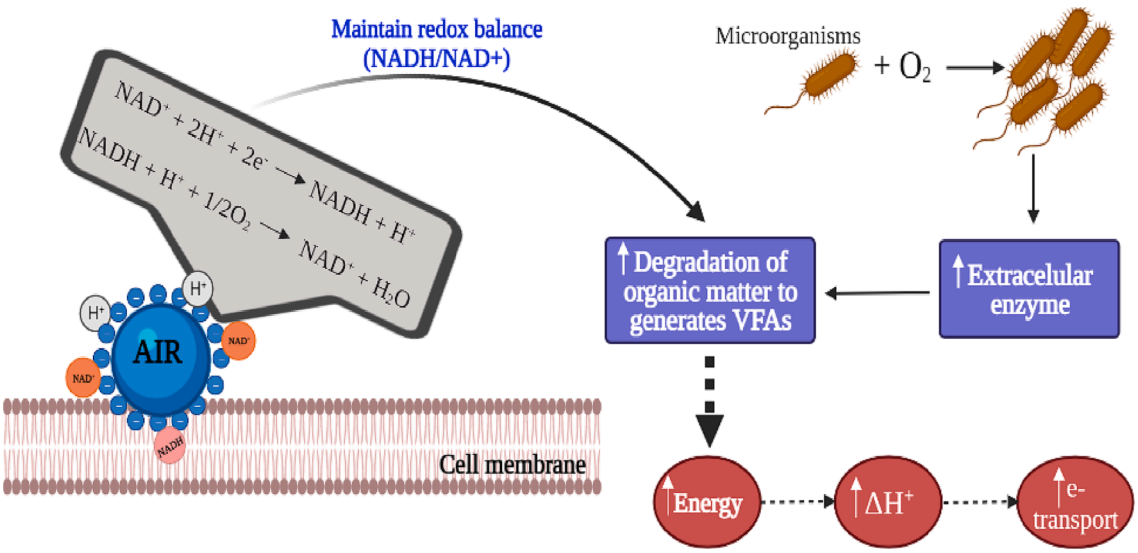


Fig. 1. Scheme of the mechanism proposed to explain microaeration effect on hydrolysis phase adapted from Zhang et al. (2021) and Zhu et al. (2022a).

Table 2
Summary of enzymatic activities reported with and without micro-aeration pretreatment.

Substrate	Enzymatic group	Enzymatic type	Activity without micro-aeration	Activity with micro-aeration	Method used	References
Food waste	Phosphatases	Alkaline phosphatase Acid phosphatase Naphthol-AS-BI-phosphohydrolase	5	15	ApiZyme Kit	Fisgativa et al. (2018)
	Lipase	Esterase (C 4) Esterase Lipase (C 8) Lipase (C 14)	0	5		
	Amino-Peptidase	Leucine arylamidase Valine arylamidase Cystine arylamidase	0	5		
	Proteases	Trypsin α-chymotrypsin	0	2		
	Glycosyl-hydrolases	α-galactosidase ß-galactosidase ß-glucuronidase α-glucosidase ß-glucosidase N-acetyl-ß-glucosaminidase α-mannosidase α-fucosidase	5	19		
Organic fraction of MSW	Cellulase	∝-galactosidase ß-galactosidase ß-glucosaminidase	8	12	ApiZyme Kit	Charles et al. (2009)
	Protease	Trypsine α-chymotrypsin	2	7		
Slugde	Protease Glycosyl-hydrolases	N/A α-glucosidase	1.5 mU/g 3.7 U/g	4.0 mU/g 4.8 U/g	Elisa Kit	Ruan et al. (2019)
Simulated food waste	Glycosyl-hydrolases	α-glucosidase	175 IU/L	275 IU/L	Elisa Kit	Li et al. (2022)
	Transferases Oxireductases	Acetate kinase Catalase	120 U/L 30 U/mL	170 U/L 40 U/mL		
Raw mixed and primary sludge	Lipase	N/A	11 Umol/mL	12 Umol/mL	Oil-alcochol emulsion	Rashvanlou et al. (2021)

3.4. Pilot and full-scale micro-aeration pretreatment applications

To date, most of the studies found in the literature related to the effect of micro-aeration pretreatment in the AD process have been limited to the laboratory scale. Several studies reported are aimed to reduce the hydrogen sulfide present in the biogas inside the anaerobic

digester as micro-aeration is applied during the AD process and not as a pretreatment process (Giordano et al., 2019; Jeníček et al., 2017). Some technologies at full scale have implemented aerobic pretreatment with different goals. For example, a technology developed by Zero Waste Energy LLC (USA) shortly aerated the waste to generate heat for the following solid stage of AD. For that, the organic waste is subjected to 12 h of force aeration to reach between 50 and 55 °C before the dry AD process. This technology has been implemented for a capacity of 90,000 tons per year under a brand of Smartferm® (Bremond et al., 2018). Another German technology called MYT™ uses aeration to reduce and homogenize the organic fraction of municipal solid waste by mixing the waste with water for 3 days at 35 °C in a plug-flow reactor to homogenize and solubilized the organic matter before AD. This technology has been used in Germany since 2006 for a treatment capacity of 100,000 tons per year (Bremond et al., 2018). Another study carried out by Jiang et al. (2018) reported the advantages of inserting a micro-aerobic tank and clarifier before a conventional activated sludge process in China with an industrial wastewater treatment capacity of 20,000 m^3 per day (reactor volume of ~11,000 m^3). The authors observed efficient pollutant removal and low sludge production (0.075 gSS/gCOD). There are also a few patent technologies related to micro-aeration in anaerobic digesters, such as patent US 10377651B2 called "Bubble generation for aeration and other purposes", and patent CN103086512B called "Method for controlling anaerobic digestion of easily-degradable organic wastes through utilizing intermittent micro-aeration". Both inventions allow the degradation of the organic matter inside the anaerobic reactor. Nevertheless, more studies must be implemented in this area to promote and expand the use of micro-aeration pretreatment strategy at the industrial level.

4. Modeling of the microaerobic process

Empirical mathematical models have been reported for microaeration processes, which generally relate the CH_4 (or biogas) production and/or solid biodegradation with the operational parameters of the microaeration process. One of the first efforts was made by Montalvo et al. (2016a), who proposed a first-order kinetic (Eq. (1.1)) for solid biodegradation:

$$\ln(VSS)_t = -k \cdot t + \ln(VSS)_0 \quad (1.1)$$

Where *VSS* is the concentration of volatile suspended solids (g/L), and *k* is a kinetic constant (h^{-1}). To validate the model, batch biodigesters were used on a laboratory scale, treating primary and secondary sludge under microaerobic conditions. The model allowed us to predict the behaviour of the experimental results, showing that the VSS solubilization can be modelled by first-order kinetics. Coefficient of determination (R^2) values higher than 0.95 were obtained, so the model adjustment was successful.

Other studies focused on methane production. Tsapekos et al. (2017) evaluated the effect of various operating parameters of the microaeration process on the AD of wheat straw. The objective was to determine the values of the O_2 load, the injections' repeatability (pulse) and the microaeration period to obtain the highest CH_4 production. Thus, they used a second-order polynomial function (Eq. (1.2)) to represent the methane yield in terms of microaeration variables:

$$\begin{aligned} CH_4 \text{ Yield} = {} & 246.506 + (6.505 \cdot A) + (21.998 \cdot B) - (17.779 \cdot C) \\ & - (0.908 \cdot A \cdot B) - (0.767 \cdot A \cdot C) + (6.300 \cdot B \cdot C) - (0.256 \cdot A^2) \\ & - (8.727 \cdot B^2) + (4.221 \cdot C^2) \end{aligned} \quad (1.2)$$

Where A is the O_2 load, B is the pulse, and C is the microaeration period. The quadratic model presented statistically significant results regarding the performance of experimental CH_4; therefore, it gives a good prediction of CH_4 yield under the studied conditions. However, the quadratic model cannot be used to predict CH_4 yield under other conditions or other substrates, describing only the results of the experiment in which it was applied. Fu et al. (2016) used a similar strategy, based on first-order kinetics applying the following equation:

$$P(t) = P_\infty \cdot \left[1 - e^{-k_h[t - L_P]}\right] \quad (1.3)$$

Where *P(t)* is the accumulated CH_4 production (mL/g VS), P_∞ is the potential CH_4 production (mL/g VS), k_h is the hydrolysis constant (d^{-1}), and L_P is the time of the lag phase (d). The parameter k_h represents the rate of the hydrolysis process under anaerobic conditions, a lower k_h means a higher hydrolysis rate. This study showed that the k_h value of experiments under microaeration was all lower than the control experiment (anaerobic digester without microaeration), which represented a higher hydrolysis rate under microaerobic conditions. This model was applied at laboratory-scale batch digesters that treated a cellulosic substrate, such as corn straw. Diaz et al. (2011) fit data from a batch test of anaerobic cellulose degradation, to which limited supplies of O_2 were applied, to a modified Gompertz equation (Eq. (1.4)).

$$P(t) = P_\infty \left[-\exp\left(\frac{R_m}{P_\infty}(\lambda - t) + 1\right)\right] \quad (1.4)$$

Where *P(t)* is the total cumulative CH_4 production (mL/g VS), P_∞ is the total CH_4 production potential (mL/g VS), R_m is the maximum CH_4 production rate (mL/g VS d), λ is the lag phase time (d) and t is the elapsed time (d). In this study, they demonstrated that microaeration reduces lag times, but it does not affect the maximum rate of CH_4 production (or hydrolysis rate constant). On the contrary, Zeb et al. (2022) fit a modified Gompertz equation to buffalo dung data for treated and non-treated substrates. The determined parameters of the modified Gompertz model allowed us to accurately predict the biogas potential of the process ($R^2 > 0.93$), showing the positive effect of microaeration on the biogas potential, although it was applied *in-situ* and not as a pretreatment. Recently, Zhan et al. (2023) proposed five models for studying the methane yield kinetics of AD after different microaeration strategies (*in situ* and as pretreatment). They found that the modified Gompertz model and the Cone model fitted very well ($R^2 > 0.99$) with the methane daily production.

In general, the empirical models obtained in microaeration pretreatment studies do not present a direct relationship between the methane generation improvement and the microaeration pretreatment, except for Tsapekos et al. (2017), since no parameters of the microaerobic pretreatment appear directly on the AD models. Therefore, the development of new phenomenological mathematical models that directly include the effect of microaeration on hydrolysis is still necessary.

The hydrolysis of biodegradable organic matter under aerobic conditions, not under microaerobic conditions, such as activated sludge, has been modelled by the activated sludge model (ASM) developed by the International Water Association (IWA) group. In ASM models, the O_2 concentration is involved in the activated sludge hydrolysis stages, therefore, it influences the degree of waste degradation (Morgenroth et al., 2002). ASM models could be used to simulate the hydrolysis of organic residues in which the microaeration process is applied. An integrated model of the ASM and the ADM1 could be an interesting alternative for the simulation of the microaerobic pretreatment and AD as a series process. The model could simulate the effect of microaeration on biogas quality, in addition to the effect on the AD hydrolysis stage.

5. Energetic and preliminary financial analysis

One aspect that can be a limitation or problem for the microaerobic pretreatment and that has been scarcely reported in the literature is the energy requirements of this pretreatment process, including financial analysis. The analysis of energy balance in an AD process including the microaerobic pretreatment was presented by Rubezius et al. (2020), who compared the energy obtained from poultry manure after a microaerobic pretreatment process, evaluating the results of biomass AD

with the indicator "energy value of biomass obtained at anaerobic conversion (e)". It was calculated using the biogas yield (B) and the energy value of biogas (e_b) obtained experimentally. They showed that the energy value from biomass measured in different ways (fresh matter, total solids and volatile solids) was always higher in the biomass pretreated microaerobically compared to the system without pretreatment. They also indicated that a longer duration of pretreatment is closely associated with higher energy consumption and may be unprofitable. Even though the energy balance was presented, a financial analysis was not performed. Later, Rashvanlou et al. (2021) used the net energy for comparing the performance of a system with microaerobic pretreatment-AD and another standard AD, using mixed sewage sludge. Net energy was calculated as the subtraction of energy outputs and energy inputs. Energy outputs were related to the energy given by methane combustion, while energy inputs were related to the energy spent for pre-treatment, the power needed for mixing and the energy to heat the anaerobic digester. According to this analysis, the net energy for an anaerobic digester with 23% VS destruction was 0.56 kWh/kg VS_{fed}, while the same system including the microaerobic pretreatment was 1.2 kWh/kg VS_{fed} being 114% higher energy output when the system uses the microaerobic pretreatment. Similarly to the work of Rubezius et al. (2020), a financial analysis was not presented in this work. Castillo et al. (2022) recently presented an energetic analysis of microaerobic pretreatment, comparing it to a more applied pretreatment such as thermal pretreatment, using mixed sludge as well. The process proposed in that study involved the presence of a boiler for increasing the temperature of the raw sludge, an air pump for the air injection, and an immersion heater to keep the temperature of aerobic reactors. This study concluded that the energy consumption of microaerobic pretreatment was higher than that of thermal pretreatment, although the consumption of microaerobic pretreatment can decrease by adding a thickening step because it reduces the volume of aeration and reduces the need for energy. According to Castillo et al. (2022), the main source of energy requirements was the boiler which was used to heat the microaerobic reactors from 22 °C to 35 °C, concluding that microaerobic pretreatment could be an interesting alternative only if the benefits such as higher solids removal and higher methane production are maintained or improved at high solids input, a situation that must be studied at a lab or pilot scale. Like the other works presented, the financial analysis was not presented in Castillo et al. (2022).

Other works have presented simplified financial analysis to compare the microaerobic pretreatment with the classic AD without any pretreatment. Ruan et al. (2019) for instance indicated that the total cost saved by microaeration was 0.0075 €/kg VS at a power cost of 0.13 €/kWh by comprehensively considering energy consumption of aeration, methane production enhancement and sludge reduction for the anaerobic treatment of mixed sludge. A deeper analysis was presented by Gonzalez-Gonzalez and Cuadros (2015) for the AD of olive mill wastewater (OMWW), including a microaerobic pretreatment. For a plant treating 28,338 m^3/year of OMWW and including costs associated with heating the anaerobic reactor, installation costs of the AD plant, consumption of electricity, operation and maintenance among others, the use of a microaerobic pretreatment before the AD has a net present value (millions of €) of 0.29, with 5,82 years of period of return on investment. These results indicate that the aerobic–anaerobic treatment of OMWW is economically feasible given that the Period of Return on Investment (PRI) is 5.82 years, three times less than the lifetime of the installation. Fu et al. (2018) studied five pre-treatments (alkali hydrogen peroxide pretreatment, NaOH pretreatment, HCl pretreatment, hot water pretreatment and microaerobic pretreatment) to study the effects of these processes on the AD of *Miscanthus floridulus*, including a preliminary economic analysis. The preliminary economic study was based on the economic input during pretreatment, start-up/running stage of AD (including the economic input for pretreatment and the energy needed for heating) and methane yield. They indicated that among these five pretreatment methods, only the microaerobic pretreatment group obtained a lower cost than an un-pretreated group to produce per cubic meter biomethane, with a cost 15.6 % lower than the system without pretreatment.

Chetawan et al. (2021) presented a preliminary cost-benefit evaluation to treat piggery waste using microaerobic pretreatment followed by AD. A volume of 450 m^3/d on average with TS 1% was analyzed, showing that the approximate energy gain from enhanced methane production would be 151,661 kWh/year. With a cost energy of 0.13 $US/kWh, the potential annual benefit from this improvement would be 19,906 $US/year. The approximate cost of retrofit by adding a microaerobic pretreatment tank and exhaust gas and cooling water heat recovery system with pumps and piping was 4,043 $US/year. Thus, the simple payback period is approximately 3.88 years. A similar analysis was presented by Zhou et al. (2021), who indicated that the total investment reduced by microaeration was 0.0079 €/kg $VS_{degraded}$ considering the power consumption of 0.13 €/kWh for the microaeration pretreatment.

To clarify the main costs and conditions that affect the microaerobic pretreatment use, an energy and simplified financial analysis was proposed, based on works presented by Fu et al. (2018) and Rashvanlou et al. (2021). According to Rashvanlou et al. (2021), net energy can be calculated as:

$$\text{Net energy} = \text{Energy Output}\,(E_o) - \text{Energy Inputs}\,(E_i) \tag{1.5}$$

Net energy was calculated as kWh/kg_{TS_added}. Energy Output was calculated as:

$$E_0 = Y_{CH_4} \cdot \varepsilon \cdot \lambda_m \tag{1.6}$$

where Y_{CH4} is the methane yield (m^3 CH_4/ kg TS added), ε expresses the lower heating value of methane (35,800 kJ/m^3CH_4) and, λ_m is the energy conversion factor of methane (0.9).

Energy inputs involve three aspects: (*i*) the energy required for heating during the pretreatment stage, and (*ii*) the energy required for microaeration and energy for heating during the running stage of the biological process.

The energy required for heating during pre-treatment was calculated as follows:

$$Q_{PT} = \frac{\rho \cdot V \cdot c_p \cdot (T_{final} - T_{initial})}{C_{TS} \cdot V} \tag{1.7}$$

where Q_{PT} is the energy needed for heating in the pretreatment (kJ/kg TS added), ρ is the density of the waste (kg/m^3), V is the volume of the reactor (m^3), c_p is the specific heat of the waste (kJ/ (kg * K)), T_{final} is the final temperature of the reactor (35 °C) and $T_{initial}$ is the initial temperature of the waste (22 °C).

The energy required for micro-aeration: Energy required for microaeration was calculated as:

$$E_{MA} = \frac{P \cdot t_{pretreatment}}{V \cdot C_{TS}} \tag{1.8}$$

Were E_{MA} is the energy needed for microaeration (kJ/$kg_{TS\ added}$), P is the power consumed by microaeration (kW); *treatment* is the pretreatment time (h); V is the volume of the microaerobic reactor (m^3) and C_{TS} is the total solid concentration of waste in the microaerobic reactor (kg/m^3).

The energy required for heating during the running stage of AD was calculated as follows:

$$Q_{AD} = \frac{\rho \cdot V \cdot c_p \cdot (T_{final} - T_{initial})}{C_{TS} \cdot V} \tag{1.9}$$

where Q_{AD} is the energy needed for heating in the anaerobic reactor (kJ/$kg_{TS\ added}$), ρ is the density of the waste (kg/m^3), V is the volume of the reactor (m^3), c_p is the specific heat of the waste (kJ/kg•K), T_{final} is the final temperature of the reactor (37 °C) and $T_{initial}$ is the initial temperature of the waste (35 °C). In this study, ρ and C of the system were

set as same with water, which is 1000 kg/m^3 and 4.18 kJ/(kg•K).

Thus, the net energy can be calculated for the system with microaerobic pretreatment as:

$$\text{Net Energy} = E_0 - Q_{PT} - E_{MA} - Q_{AD} \quad (1.10)$$

In the case of the system without microaerobic pretreatment:

$$\text{Net Energy} = E_0 - Q_{AD} \quad (1.11)$$

During the pretreatment and running stage, the energy needed was provided by electric energy and the energy transfer efficiency was set as 80%. The electricity price according to the literature was 0.2 €/kWh. The values used for analysis are shown in Table 3. Three parameters (solid concentration (g TS/L), time of aeration and percentage of increase of methane due to microaerobic pretreatment) were studied to see the effect on energy and financial results. Results (see supplementary materials) indicated that for an aeration time of 48 h and an increase of 50% in CH_4 with respect to the control, the microaeration has a better financial result only when the solid concentration of waste (in g TS/L) is higher than 30 g TS/L (3% of TS). Regarding the effect of aeration time, it was obtained that aeration time between 48 h and 72 h allows higher financial results using pretreatments in comparison to a system without pretreatment, in the case of a waste with 5% of solids and increment of 50% in CH_4 percentage respect to the control when microaeration is used. Finally, the percentage of CH_4 increasing superior to 30% regarding the system without pre-treatment results in a better financial performance for waste with 50 g/L of total solids and 48 h of aeration time.

Therefore, it can be indicated that wastes with solids content superior to 3%, aeration time lower than 72 h and increasing of CH_4 methane higher than 30% could allow financial viability of microaeration pretreatment.

6. Limitations and future perspectives

Although the evidence shows that the microaerobic pretreatment has been useful at the lab and pilot scale, more efforts must be done at the industrial scale. In addition, it is important to consider that microaerobic pretreatment may have limitations in its effectiveness when substrates are used where hydrolysis is not the limiting step. This is because the addition of air can stimulate bacterial growth and increase biomass production, leading to greater sludge production. Therefore, it is necessary to carefully evaluate whether the application of this technique is appropriate based on the specific characteristics of the substrate being treated. On the other hand, issues such as energy analysis and life cycle assessment have not been addressed. These aspects must be included in the analysis not only to determine if the process is better than other pretreatment technologies but also to determine which aspects of the process should be improved to obtain a green technology. Mathematical modelling of microaerobic pretreatment also required study, especially in terms of kinetic under low oxygen concentration and oxygen mass transfer, explicitly including the effect of aeration on hydrolysis and methane production. In this sense, a better understanding of the dynamic of oxygen in the bioreactor will improve the control of this parameter at the pilot and industrial scale. Furthermore, studies are required to evaluate the effect of microaereation on subsequent thermophilic and/or mesophilic AD. This aspect is of utmost importance because depending on the temperature, certain groups of microorganisms will prevail, which will impact the stability of the anaerobic process and, consequently, the biogas production.

Another limitation related to the microaeration as pretreatment is the residual oxygen that remains in the mixture to be fed into the anaerobic reactor, particularly at large scales. Due to the large volume of substrate being processed, its depletion may take longer. Residual oxygen may inhibit the activity of anaerobic microorganisms if is in excess. One economic solution to solve this issue is to guarantee the correct mixing and agitation of the substrate in the digester to distribute the oxygen uniformly and promote a faster depletion.

Finally, other aspects such as the effect of micropollutants removal in a different type of waste (sewage sludge, pig manure, agro-industrial solid waste), also should be studied to determine other characteristics of the pretreatment.

Table 3
Values used for preliminary economic analysis.

Parameters	Values	References
Volumetric flow, m^3/d	70	This work
Solid concentration, g TS/L	40	Castillo et al. (2022)
Volume of microaerobic reactor pre-treatment	70	This work
ρ, kg/m^3	1000	Fu et al. (2018)
c_p, kJ/kg K	4.18	Fu et al. (2018)
methane increasing by pretreatment, %	50	Castillo et al. (2022)
Y (m^3 CH_4/kg TS added)	450	Wehner et al. (2021)
ε, lower heating value of methane, kJ/m^3	35,800	Rashvanlou et al. (2021)
λ_m, energy conversion factor of methane	0.9	Rashvanlou et al. (2021)
η (electric efficiency blowers)	0.8	Rashvanlou et al. (2021)
Aeration time, d	2	Castillo et al. (2022)
vvm	0.35	Castillo et al. (2022)
Volumetric flow of blower (Submerged venturi type), m^3/h	200	https://www.dlc.cl/wp-content/themes/dlc/archivos/aireadores.pdf in Spanish
Power of 1 blower (Submerged venturi type), kW	9	https://www.dlc.cl/wp-content/themes/dlc/archivos/aireadores.pdf in Spanish
Price of electricity	0.2	Wehner et al. (2021)

7. Conclusions

Microaerobic pretreatment has shown good performance for increasing biogas volume at the lab, pilot and industrial scale, with improvement from 30% up to 216%. The pathway behind the process shows that oxygen affects redox balance NADH/NAD^+, driving the generation of the enzymes by acidogenic biomass and improving the hydrolysis rate. The energy balance and preliminary financial analysis showed that wastes with solids content superior to 3%, aeration time lower than 72 h and an increase of CH_4 higher than 30% allow financial viability of microaeration pretreatment. More studies are necessary for aspects such as life cycle assessment and mathematical modelling.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

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